

PhishGuard Final Research Report

Project Overview

PhishGuard is a cybersecurity focused web application concept designed to help users identify suspicious emails, text messages, and URLs through explainable phishing risk analysis and cybersecurity awareness guidance. The project is currently in the Capstone 1 planning and documentation phase, with the possibility of a lightweight prototype depending on team bandwidth.

The purpose of this research report is to organize the nine selected sources into a clear foundation for the project. These sources help support PhishGuard's scope by explaining common phishing warning signs, user awareness needs, current phishing trends, explainable analysis, risk scoring, secure URL handling, and real world phishing examples.

Research Purpose

The research was compiled to answer three main questions for the team.

- First, what types of phishing risks should PhishGuard help users recognize?
- Second, how can PhishGuard explain suspicious content in a way regular users can understand?
- Third, how can the project stay realistic for Capstone 1 while still being useful, current, and cybersecurity focused?

The nine sources show that PhishGuard should not only identify suspicious content, but also explain why the content may be risky and guide the user toward safer digital decisions.

Source 1: CISA, Recognize and Report Phishing

CISA's source supports the foundation of PhishGuard's phishing warning system. It explains that suspicious messages may include harmful links, emails, or attachments that try to collect personal information or infect devices. This connects directly to PhishGuard because the application is meant to help users recognize suspicious content before they click links, open attachments, or share sensitive information. ([CISA][1])

For the project, this source supports warning categories such as suspicious links, unexpected attachments, personal information requests, and suspicious messages. These categories can help shape PhishGuard's rule based analysis and risk explanation system.

This source is most useful for the problem statement, scope statement, risk assessment, and phishing indicator documentation.

Source 2: FTC, How To Recognize and Avoid Phishing Scams 🐟

The FTC source supports PhishGuard's user awareness and safety guidance features. It explains that scammers use emails and text messages to trick users into giving away personal or financial information. It also provides reporting guidance for phishing emails and phishing text messages. ([Consumer Advice][2])

This source is important because PhishGuard is not only meant to flag suspicious content. It should also help users understand what to do next. The FTC supports user recommendations such as not clicking suspicious links, not sharing personal information, verifying the sender, and reporting suspicious messages.

This source is especially useful for the user guidance section of the project, including possible app messages, safety recommendations, and acceptance criteria.

Source 3: APWG, Phishing Activity Trends Report, Q4 2025 🐟

The APWG report supports the current phishing trend side of PhishGuard. The Q4 2025 report states that SMS phishing increased during that quarter, while QR code phishing decreased by 9 percent. APWG also tracks phishing activity through reported phishing emails and other data from contributing members. ([docs.apwg.org][3])

This source helps justify why PhishGuard's scope includes more than email. Since the project includes suspicious emails, text messages, and URLs, APWG supports the need to consider modern phishing methods such as SMS phishing, impersonation, and URL based attacks.

This source is useful for the problem statement, project justification, risk assessment, and poster presentation.

Source 4: Explainable Phishing Website Detection for Secure and Transparent Online Transactions 🐟

This source supports PhishGuard's explainable risk analysis goal. The article explains that phishing websites often depend on misleading URLs to trick victims and identifies phishing sites as a major attack vector for online fraud. ([PMC][4])

This connects strongly to PhishGuard because the project is not simply about telling users that something is suspicious. The goal is to explain why a message, text, or URL may be risky. If PhishGuard flags a suspicious URL, the user should be able to see which warning signs contributed to the result.

This source supports the explainable risk score, URL analysis, and future risk scoring engine. It also helps justify why the app should give users plain language explanations instead of only showing a low, medium, or high risk label.

Source 5: Exploring the Evidence for Email Phishing Training 🐟

This source supports the cybersecurity awareness and education side of PhishGuard. The article is a scoping review that examined 42 studies related to phishing training interventions and measurable outcomes on user susceptibility. ([ScienceDirect][5])

This source is useful because PhishGuard is not just a scanner. It is also an awareness tool. The project should help users learn from the warning signs found in suspicious content. The research supports the idea that user education matters, but users still benefit from clear guidance when they face real phishing attempts.

For PhishGuard, this source supports plain language explanations, educational feedback, and user centered guidance. It can also support user stories and acceptance criteria related to cybersecurity awareness.

Source 6: Microsoft Digital Defense Report 2025 🐟

Microsoft's 2025 report supports the modern threat landscape section of the project. Microsoft Incident Response found that 28 percent of breaches were initiated through phishing or social engineering. The report also mentions newer methods such as ClickFix social engineering and device code phishing. ([Microsoft][6])

This source helps show why PhishGuard is relevant. Phishing and social engineering remain major real world cybersecurity threats. This strengthens the project's problem statement and helps explain why users need support before clicking links, entering credentials, or trusting suspicious messages.

For the project, this source supports the overall need for PhishGuard as a safer digital decision making tool.

Source 7: NIST Phish Scale User Guide

The NIST Phish Scale User Guide supports PhishGuard's risk scoring and evaluation framework. NIST explains that the Phish Scale is a method created to help cybersecurity awareness training implementers rate how difficult a phishing email is for users to detect. The guide includes instructions for applying the scale to phishing emails. ([NIST][7])

This source is important because PhishGuard's tentative technical direction includes rule based phishing detection logic and a risk scoring engine. NIST gives the team a structured way to think about phishing difficulty, user detection, cues, and context.

For PhishGuard, this source can support the design of risk levels. Instead of assigning random scores, the project can explain that different warning signs contribute to the final result. For example, suspicious links, sender mismatch, urgent language, and requests for sensitive information could all influence the risk level.

This source is useful for the risk scoring engine, acceptance criteria, and possible prototype logic.

Source 8: OWASP, Testing for Client Side URL Redirect

OWASP's source supports the secure web application side of PhishGuard. It explains that client side URL redirection can become an input validation problem when an application accepts user controlled input that leads to an external URL. OWASP notes that this kind of vulnerability can be used for phishing or redirecting a victim to an infection page. ([OWASP][8])

This source is different from the general phishing awareness sources because it focuses on how the PhishGuard application itself should handle URLs safely. Since PhishGuard may allow users to submit suspicious URLs, the team should think carefully about how those links are displayed, stored, or processed.

For example, PhishGuard should avoid automatically opening submitted URLs. It should also avoid making dangerous links too easy for users to click. If a lightweight prototype is created, this source can help guide secure input handling and URL display decisions.

This source is useful for the technical direction, risk assessment, and prototype security considerations.

Source 9: Cloudflare, The Mechanics of a Sophisticated Phishing Scam and How We Stopped It 🐟

Cloudflare's case study supports the real world phishing example section of the project. The article describes a targeted SMS phishing attack with similar characteristics to the attack against Twilio. Cloudflare explains that its team blocked the phishing domain using Cloudflare Gateway and responded to the incident through several security actions. ([The Cloudflare Blog][9])

This source is valuable because it gives the project a concrete phishing case. PhishGuard's scope includes text messages and URLs, so a real SMS phishing case helps show why the project should not focus only on email. It also connects to the idea that phishing attacks often combine social engineering, links, fake login pages, and urgency.

For PhishGuard, this source supports SMS phishing awareness, URL risk analysis, and safer user decision making. It is especially useful for the problem statement, presentation, and project poster because it gives the team a real example to explain why the project matters.

Overall Research Findings 🐟

The nine sources support PhishGuard from several angles.

CISA and the FTC support the basic warning signs and user safety guidance. These sources help explain what users should watch for and what they should do when they receive suspicious emails, texts, or links.

APWG and Microsoft support the current threat landscape. They show that phishing, SMS phishing, impersonation, and social engineering remain active cybersecurity concerns.

The explainable phishing detection article and NIST Phish Scale support the risk analysis side of the project. These sources help justify why PhishGuard should explain risk and use structured warning indicators instead of giving users a vague result.

The phishing training article supports the educational purpose of PhishGuard. It shows that awareness and training are important, but users still need practical support when facing suspicious content.

OWASP supports secure web application planning. Since PhishGuard may process user submitted URLs, the team should consider how to safely handle links in a possible prototype.

Cloudflare provides a real world case study. It gives the team a concrete example of SMS phishing and URL based social engineering, which fits PhishGuard's scope well.

Connection to PhishGuard Features

The research supports several possible PhishGuard features:

Suspicious Content Submission

Users should be able to submit an email, text message, or URL for review. This feature is supported by CISA, FTC, APWG, Cloudflare, and the explainable phishing detection source because phishing can appear through multiple channels.

Warning Sign Detection

PhishGuard should check for warning signs such as suspicious links, unexpected attachments, requests for personal or financial information, impersonation, urgent language, and SMS phishing. CISA, FTC, APWG, Microsoft, and Cloudflare all support this part of the project.

Explainable Risk Score

PhishGuard should provide a risk level such as low, medium, or high. More importantly, the app should explain what caused that risk level. The explainable phishing detection source and NIST Phish Scale support this feature.

User Safety Guidance

After identifying suspicious indicators, PhishGuard should recommend safer actions. Examples include not clicking links, not opening attachments, not replying with private information, verifying the sender through an official website, and reporting suspicious messages. CISA and the FTC strongly support this part of the project.

Cybersecurity Awareness Support

PhishGuard should help users learn from each analysis. The phishing training source supports the idea that education and feedback can help users become more aware of phishing risks.

Secure URL Handling

If the team creates a prototype, PhishGuard should handle submitted URLs carefully. OWASP supports this part of the project by showing how unsafe redirects and untrusted URL input can become phishing risks.

Recommended Risk Categories for PhishGuard

Based on the research, PhishGuard can organize its analysis into these categories:

Link Risk

This category checks whether the submitted content includes a suspicious, shortened, misleading, or unexpected URL.

Attachment Risk

This category checks whether the message includes an unexpected attachment or asks the user to download a file.

Information Request Risk

This category checks whether the message asks for passwords, financial information, account details, verification codes, or other sensitive information.

Impersonation Risk

This category checks whether the message appears to pretend to be from a trusted company, school, bank, delivery service, platform, or organization.

Urgency Risk

This category checks whether the message pressures the user to act quickly, such as claiming that an account will be closed, a payment is overdue, or immediate action is required.

SMS Phishing Risk

This category checks whether the suspicious content comes through a text message and includes links, impersonation, or requests for personal information.

URL Handling Risk

This category applies to PhishGuard's own web application design. It reminds the team that submitted URLs should be handled safely and should not create unnecessary risks for users.

Suggested Result Format for PhishGuard

A possible PhishGuard result could include the following sections.

Risk Level: High

Warning Signs Found: Suspicious link, request for personal information, urgent language

Explanation: This message may be risky because it asks the user to click a link and provide sensitive information quickly. Phishing scams often use links, text messages, and social engineering tactics to trick users into revealing private information.

Recommended Action: Do not click the link. Do not provide personal information. Visit the official website directly or contact the organization through a trusted method.

Educational Note: Messages that pressure users to act quickly or provide sensitive information should be treated carefully.

This type of result supports PhishGuard's main goal because it identifies risk, explains the reason, and guides the user toward a safer action.

How the Research Supports Team Roles

- For the Product Owner, the research strengthens the project vision by showing why PhishGuard is useful and relevant.
- For the Team Lead and Scrum Coordinator, the research can help organize future tasks, such as creating a risk category chart, refining the backlog, and planning documentation updates.
- For Research and Documentation, the research provides evidence for the problem statement, scope statement, risk assessment, and final report.
- For Design and UX, the research supports simple warnings, plain language explanations, and clear user guidance.
- For QA and Requirements, the research supports acceptance criteria, such as requiring the system to identify suspicious links, display warning categories, and provide user friendly explanations.

Suggested Acceptance Criteria Based on Research

- The system should allow users to submit suspicious email, text message, or URL content for review.
- The system should identify common phishing warning signs such as suspicious links, personal information requests, urgent language, and impersonation.
- The system should provide a risk level based on the warning signs found.
- The system should explain why the submitted content may be suspicious.
- The system should provide at least one recommended safety action.
- The system should use plain language so non expert users can understand the result.
- The system should avoid automatically opening submitted URLs.

- The system should keep the analysis focused on awareness and explainable risk guidance.

Final Conclusion

The nine research sources provide a strong foundation for PhishGuard. Together, they show that phishing remains a relevant cybersecurity issue across emails, text messages, URLs, impersonation attempts, and social engineering attacks. They also support the project's focus on awareness, explainability, and safer digital decision making.

The research suggests that PhishGuard should remain realistic for Capstone 1 by focusing on clear warning categories, simple risk levels, and plain language explanations. The project does not need to become a fully advanced enterprise security system. Its value comes from helping users pause, understand suspicious content, and make safer choices before clicking links, opening attachments, or sharing sensitive information.

Overall, the research supports PhishGuard as an explainable phishing awareness tool with a practical and manageable scope.

Work Cited

- Anti-Phishing Working Group. Phishing Activity Trends Report, 4th Quarter 2025. APWG, 18 Feb. 2026, docs.apwg.org/reports/apwg_trends_report_q4_2025.pdf.
- Cloudflare. “The Mechanics of a Sophisticated Phishing Scam and How We Stopped It.” The Cloudflare Blog, 9 Aug. 2022, blog.cloudflare.com/2022-07-sms-phishing-attacks/.
- Cybersecurity and Infrastructure Security Agency. “Recognize and Report Phishing.” CISA, www.cisa.gov/secure-our-world/recognize-and-report-phishing.
- Dawkins, Shantee, and Jody Jacobs. NIST Phish Scale User Guide. National Institute of Standards and Technology, 15 Nov. 2023, doi.org/10.6028/NIST.TN.2276.
- Federal Trade Commission. “How To Recognize and Avoid Phishing Scams.” Consumer Advice, Federal Trade Commission, consumer.ftc.gov/articles/how-recognize-avoid-phishing-scams.
- Kehkashan, Tanzila, et al. “Explainable Phishing Website Detection for Secure and Sustainable Cyber Infrastructure.” Scientific Reports, vol. 15, article 41751, 2025, doi.org/10.1038/s41598-025-27984-w.
- Marshall, Nina, Daniel Sturman, and Jaime C. Auton. “Exploring the Evidence for Email Phishing Training: A Scoping Review.” Computers & Security, vol. 139, 2024, article 103695, doi.org/10.1016/j.cose.2023.103695.
- Microsoft. Microsoft Digital Defense Report 2025. Microsoft, 2025, www.microsoft.com/en-us/corporate-responsibility/cybersecurity/microsoft-digital-defense-report-2025/.
- OWASP Foundation. “Testing for Client-Side URL Redirect.” OWASP Web Security Testing Guide, OWASP Foundation, owasp.org/www-project-web-security-testing-guide/latest/4-Web_Application_Security_Testing/11-Client-side_Testing/04-Testing_for_Client-side_URL_Redirect.